

Nextflow run introduction



How to run a workflow using nextflow command

Run your first nextflow pipeline

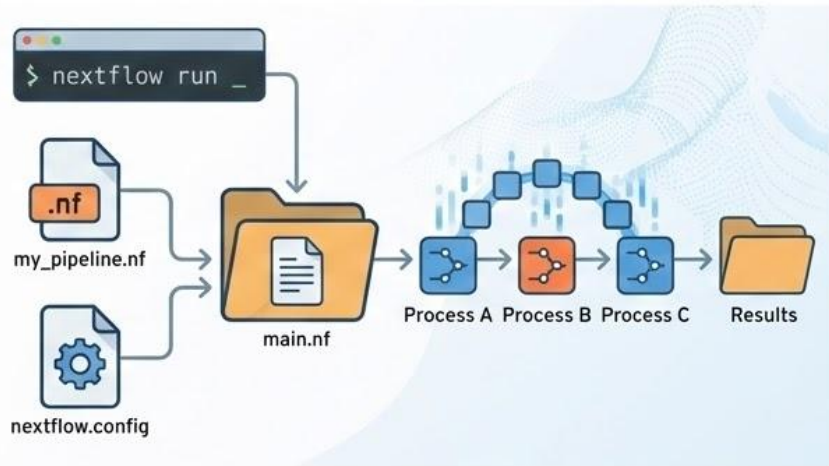
Materials: `practicals/day1/1-nextflow_run_introduction`

- At the barely minimum, a Nextflow file is composed by a **single file with extension .nf**
- Often a configuration file is also included named *nextflow.config*
- If the pipeline file is named *main.nf* it will be recognized automatically by nextflow
- You run a pipeline by using the **nextflow run command pointing to a valid .nf file** or to a folder containing a *main.nf* file
- You should execute each pipeline from a dedicated working directory (Nextflow will create dedicated files, your results and various logs) ⇒ You can repeat / resume the same pipeline in the same working directory, but **it's not recommended to run different pipelines from the same working directory.**

Run your first nextflow pipeline

Materials: [practicals/day1/1-nextflow_run_introduction](#)

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- Often a configuration file is also included named `nextflow.config`
- If the pipeline file is named `main.nf` it will be recognized automatically by nextflow
- You run a pipeline by using the `nextflow run` command pointing to a valid `.nf` file or to a folder containing a `main.nf` file



Run your first nextflow pipeline

Code_example_1

- hello-nextflow.nf
- nextflow.config



→ Nextflow run path/to/code_example_1/hello-nextflow.nf



Code_example_1

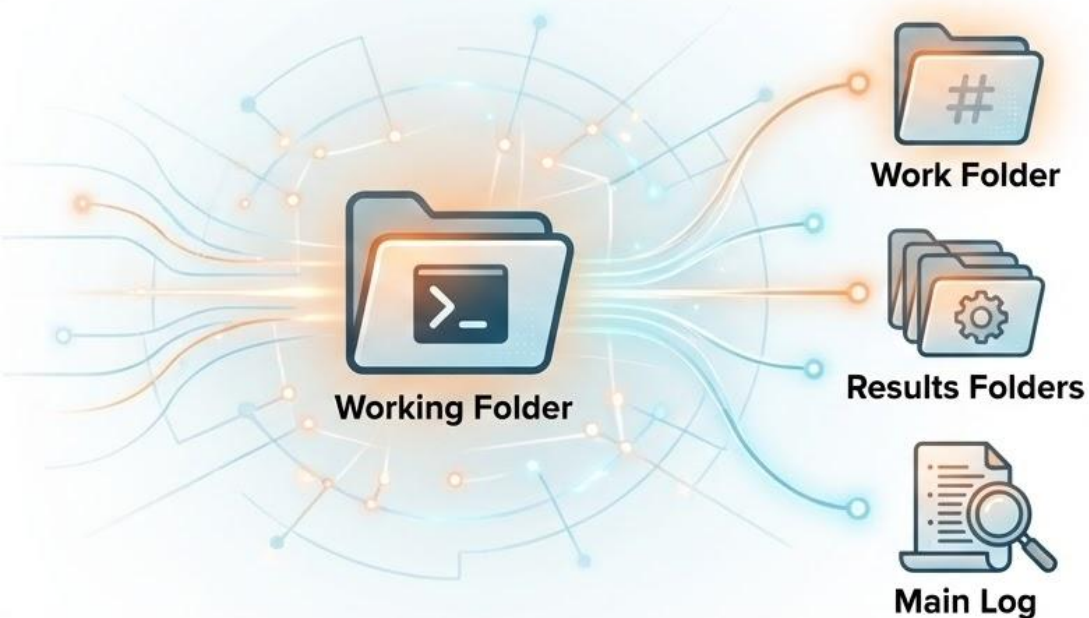
- Code_example_1
 - main.nf
 - nextflow.config

→ Nextflow run path/to/code_example_1



Understand a pipeline output

By default, during the execution the pipeline will generate data in the working folder (the folder from where nextflow run is executed).



A work folder is always generated and contains all the data of the single operations.

One or more results folders may be created if configured in the pipeline.

The main log allows to monitor pipeline execution and see the single processes that are performed.

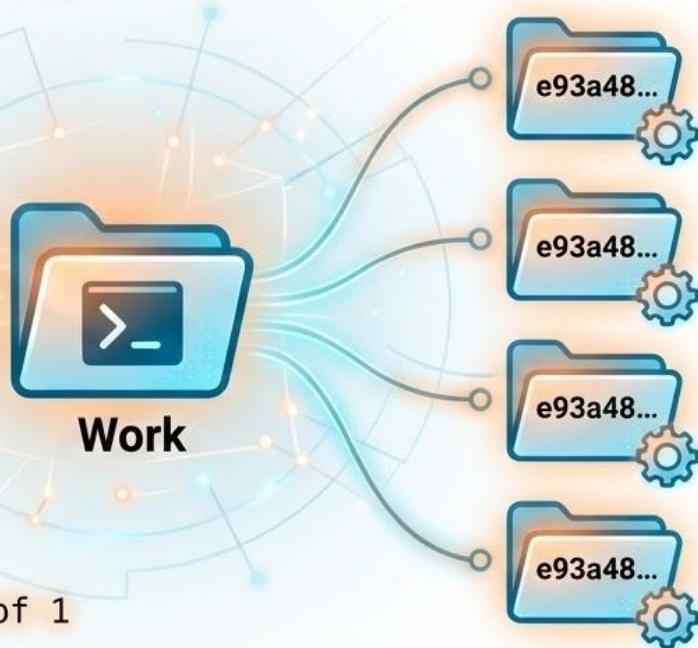
The work folder

When you run a workflow, Nextflow creates a distinct 'task directory' for every single invocation of each process...



If you look at the log of the previous execution, you will see something like this:

```
[1c/e93a48] sayHello | 1 of 1
```



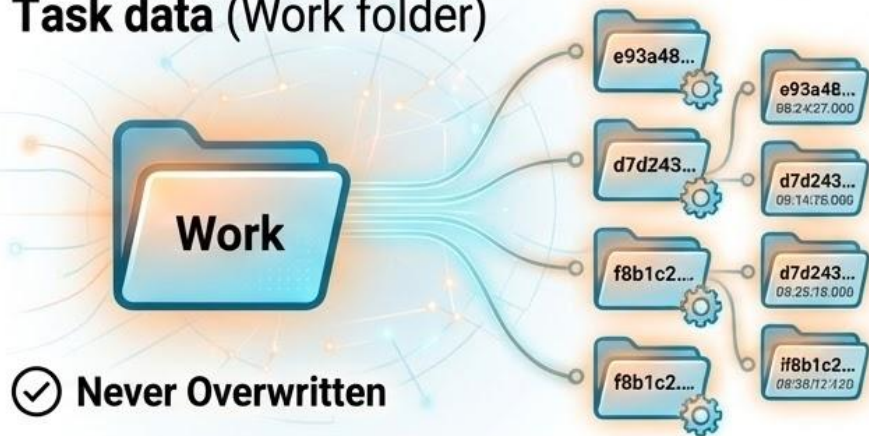
Work

```
|--1c
|   |--e93a488bf...
|   |--...
|--4d
    |--g32a872te...
    |--...
```


Task data vs results data

Nextflow creates a new task directory under the work folder, with a different hash string, for each task execution across runs.

Task data (Work folder)



✓ **Never Overwritten**

Hence, files in the work folder are never overwritten when you run the a pipeline from the same working folder multiple times.

Results data (Results folder)

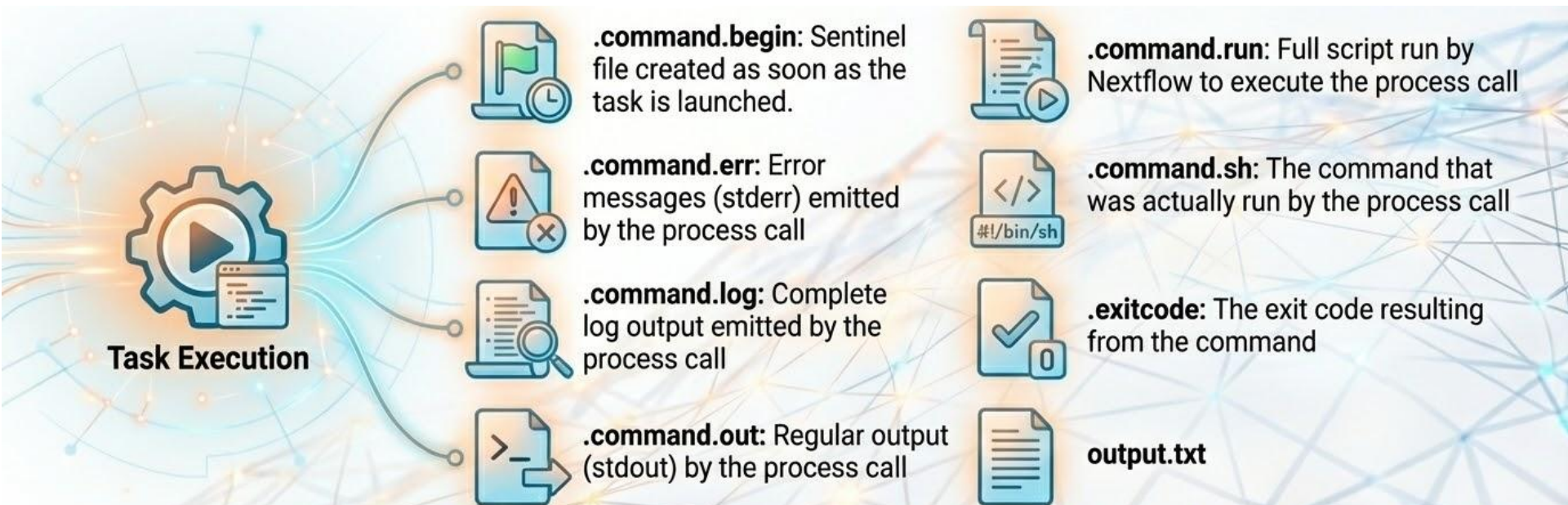


↻ **Overwritten**

On the other hand, the final result file results/output.txt is overwritten with the new output.

Helper files generated by Nextflow for a task

For all task executions, Nextflow creates the following helper and log files in the task directory



Basis of Nextflow CLI



Nextflow run CLI arguments

- Nextflow run options customize pipeline execution behavior
- **Double dash (--) options** specify pipeline script parameters and variables
- **Single dash (-) options** modify how Nextflow itself runs (set work directory, add config files, set profiles, etc.)

nf

```
nextflow run
--r 3.10.1
--input samplesheet.csv
--genome GRCh38
--genome GRCh38
--fq_tool fastqc
--max_cpus 8
--max_memory "64.GB"
--max_memory "64.GB"
--profile docker
```

```
NE XT FLOW <
NE XT FLOW <
```

version 23.04-0-edge

```
ing "nf-core/rnaseq:[amazing_euclid] - revisiaff
> Local [1]
```

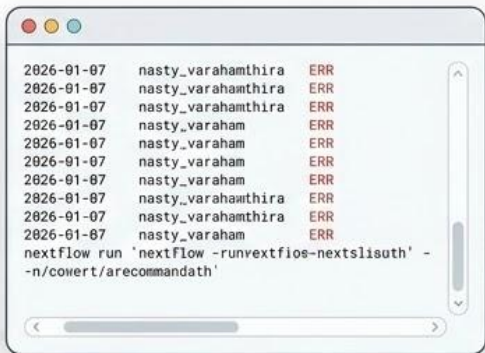
Useful nextflow run options

- Specify parameters using the command line:
 - use **--message** to specify a custom value for the message pipeline parameter.
- Specify a custom work directory:
 - The **-work-dir** option sets a custom work folder location.
- Complement/override configuration:
 - Use **-c** to pass additional configuration files



Log and clean

Inspect previous runs with ``nextflow log``



View Execution History

Display a list of previous runs, including their unique run names, start and end times, status, and the command used.

```
nextflow log
```

Remove intermediate data with ``nextflow clean``



Manage Disk Space

Clean intermediate files generated by task executions of a previous run. Use `-n` for dry-run, `-f` for permanent deletion.

```
nextflow clean -n <run_name>  
nextflow clean -f <run_name>
```